

DISTRIBUTED SYSTEMS
Principles and Paradigms
Second Edition
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Chapter 1

Introduction

Definition of a Distributed System (1)

- A distributed system is:
- A collection of independent computers that appears to its users as a single coherent system.

Definition of a Distributed System (2)

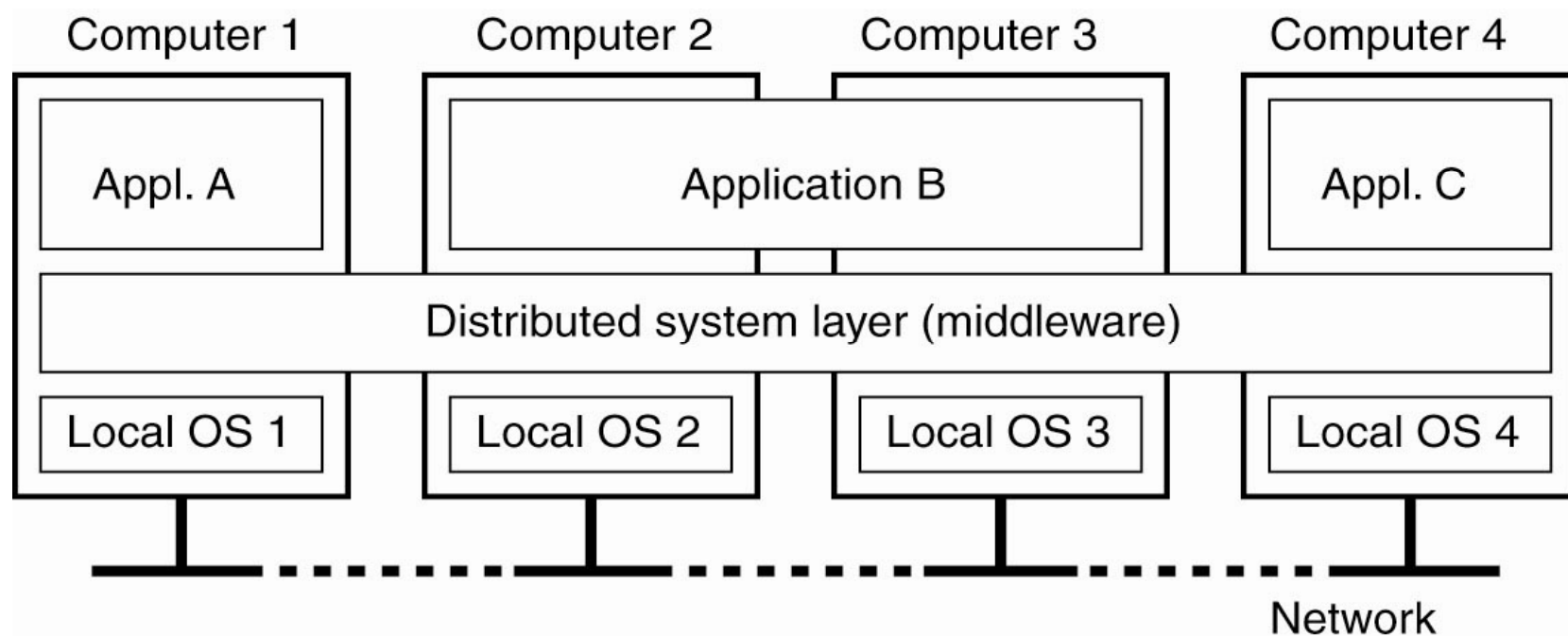


Figure 1-1. A distributed system organized as middleware. The middleware layer extends over multiple machines, and offers each application the same interface.

Goals Of D.S.

Make
Resources
Easily
Accessible.

Hide the
fact that
resources
are
distributed
across a
network.

It should
be Open.

It should
be
Scalable.

Making Resources Accessible

Main goal of D. S. is to make it easy for the users to access remote resources and share them.

Reasons for sharing Resources:

- Economics
- Collaborate and Exchange Information.

Security.

Transparency in a Distributed System

Transparency	Description
Access	Hide differences in data representation and how a resource is accessed
Location	Hide where a resource is located
Migration	Hide that a resource may move to another location
Relocation	Hide that a resource may be moved to another location while in use
Replication	Hide that a resource is replicated
Concurrency	Hide that a resource may be shared by several competitive users
Failure	Hide the failure and recovery of a resource

Figure 1-2. Different forms of transparency in a distributed system (ISO, 1995).

Degree of Transparency

- Trade-off between high degree of transparency and the performance of a system.
- Conclusion:
 - Aiming for distribution transparency is a nice goal.
 - Should be considered with other issues:
 - Performance.
 - Comprehensibility.
 - Price for not being full transparent may be **high**

Openness

- Is a system that offers services according to standard rules that describe the syntax and semantics of those services.
 - Interface Definition Language.
 - Interoperability.
 - Portability.
 - Extensible.

Opnness (Cont.)

- Separating Policy from Mechanism:
 - System should be as a collection of relatively small and easily replaceable or adaptable components.
 - E.g. monolithic approach (closed system).
 - World wide web (Caching).

Scalability

Can be measured along at least three different dimensions:

With respect to size (add users, resources).

Geographically scalable.

Administratively Scalable.

Scalability Problems (size)

Concept

- Centralized Services
- Centralized Data
- Centralized Algorithms

Examples

- A single server for all users.
- A single on-line telephone book.
- Doing routing based on complete information

Scalability Problems

Characteristics of decentralized algorithms:

- No machine has complete information about the system state.
- Machines make decisions based only on local information.
- Failure of one machine does not ruin the algorithm.
- There is no implicit assumption that a global clock exists.